

Market Structures and the Distribution of Content

Matthew J. H. Murphy
University of Pennsylvania
July 22, 2026

Motivation

Entertainment products and their markets are unique:

Product characteristics:

- Both horizontal (genre) and vertical (quality) differentiation.
- Quality is ex-ante uncertain.

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- The main result of the paper is that these frictions **jointly** lead to a market inefficiency.

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These markets are also unique:

- Historically, significant pricing frictions.
- **Digitalization has radically changed these markets.**

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- The main result of the paper is that these frictions **jointly** lead to a market inefficiency.

Model

Products and Producers

Unit mass of producers.

- Producers have idiosyncratic quality $v \sim F$ (independently realized).
- Each producer chooses to produce in a specific genre $g \in G = \{1, \dots, G\}$.
 - A producer in genre g produces a good of quality equal to their quality v .

Assumption

Let $\bar{F}(v) = 1 - F(v)$. F has increasing hazard rates: $\frac{f(v)}{\bar{F}(v)}$ is strictly increasing.

Two settings: producers observe quality either before or after genre choices.

Whether or not producers observe their quality before they make production changes the strategy space.

Ex-ante Realization of Quality.

A population strategy for producers corresponds to a mapping $\Psi : \mathbb{R} \rightarrow \Delta G$, where $\Psi(v)$ is the (possibly mixed) strategy of a producer with quality v .

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Ex-post Realization of Quality.

A strategy for producers corresponds to a mapping $\Psi : [0, 1] \rightarrow \Delta G$.

- Ψ induces a genre allocation $\psi \in \Delta G$ of producers: $\psi_g = \int_0^1 \Psi_g(i) di$.
- ψ induces μ^ψ the measure of content in genre g : $\mu_g^\psi((v, \infty)) = \psi_g \cdot \bar{F}(v)$.

There is a unit mass of consumers.

- Each consumer has a type (g, t) :
 - g determines which genre he consumes.
 - t determines *how much* content he can consume.
- Let T be the distribution of types.
- Let T_g be the distribution of demand for genre g , derived from T .

Assumption

T_g admits a continuous density $\frac{dT_g}{dt}(t)$ away from 0, and is full-support on $[0, \bar{t}_g]$.

Consumer Utility and Prices

Consumers receive utility proportional to the quality of consumed goods.

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Consumers also pay for each good that they consume. Like with producer information, focus on two pricing regimes:

- **Competitive Pricing:** a producer of quality v chooses a price in genre g , $p_g(v)$, to maximize profits.
- **Fixed, Exogenous Pricing:** there is a single market price $p \equiv p_g(v)$ for all goods and genres.

The net utility for a consumer is hence $(v - p_g(v))dv$ for each product of quality v consumed.

Regimes and Timing.

There are four possible regimes.

- Competitive vs fixed pricing.
- Ex-post vs ex-ante realization of quality.

The timing depends on the market setting. Generally:

- Producers choose genres, qualities are realized (order depends on the setting).
- Prices are chosen, observing competition.
- Consumers choose bundles.
- Consumer utility and producer profits are realized.

Main Result

Main Result: Both frictions are needed for market failure

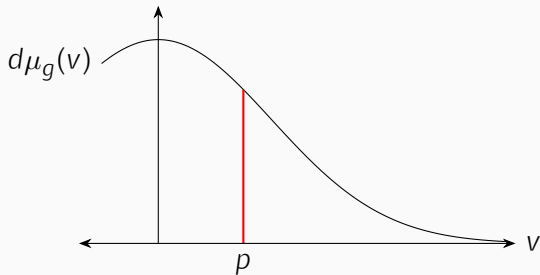
	Ex-ante Realization of Quality	Ex-post Realization of Quality
Competitive Pricing	No Inefficiency Explanation	No Inefficiency Explanation
Fixed Pricing	No Inefficiency* Explanation	Allocative Inefficiency

*There is an inefficiency stemming from positive value products not being consumed: a “Posted Price” inefficiency.

Joint impact of frictions.

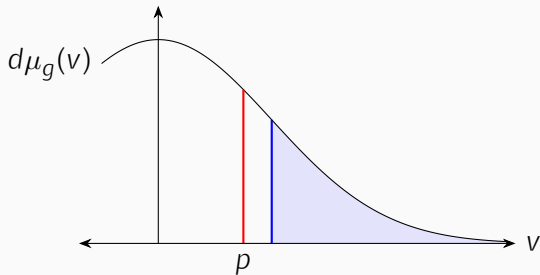
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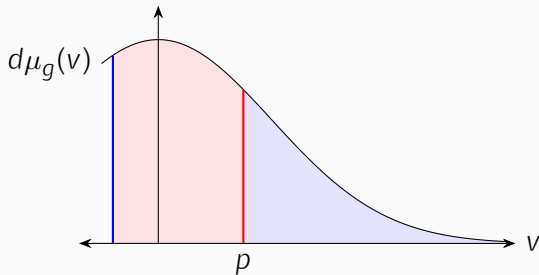
$$\int_{v_g(t)}^{\infty} (v - p) d\mu_g(v)$$

where $v_g(t)$ is such that

$$\mu_g([v_g(t), \infty)) = t$$

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$$\mu_g([v_g(t), \infty)) = t$$

Producer Profits

Fix an allocation of producers $\psi \in \Delta G$ and $p \geq 0$.

For a consumer (g, t) , face content μ_g^ψ .

- Threshold $v_g(t)$ will solve $t = \psi_g \bar{F}(v)$.

Hence, will consume everything above $\max \left\{ p, \bar{F}^{-1} \left(\frac{t}{\psi_g} \right) \right\}$.

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Facing competition ψ , expected profits in genre g are given by

$$\int_0^\infty \Pi_p \left(\min \left\{ \bar{F}(p), \frac{t}{\psi_g} \right\} \right) dT_g(t), \quad \text{where} \quad \Pi_p(x) := \int_{\bar{F}^{-1}(x)}^\infty p f(v) dv$$

For $p = 0$, let $\Pi_0(x) := \int_{\bar{F}^{-1}(x)}^\infty f(v) dv$.

Definition

An equilibrium ψ^p for market price p is an allocation of producers such that

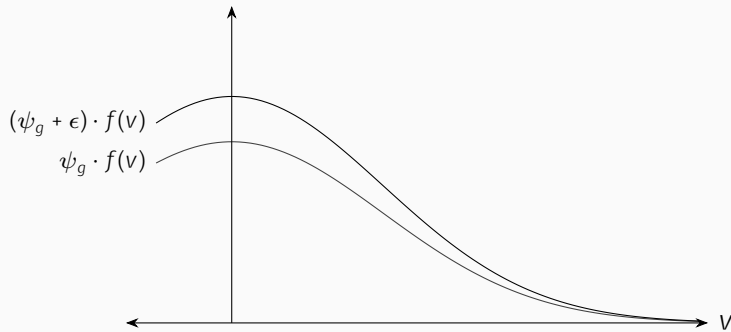
$$\psi_g^p > 0 \Rightarrow g \in \operatorname{argmax}_{g \in G} \int_0^\infty \Pi_p \left(\min \left\{ \bar{F}(p), \frac{t}{\psi_g^p} \right\} \right) dT_g(t)$$

Lemma

ψ^p is unique for each $p \geq 0$.

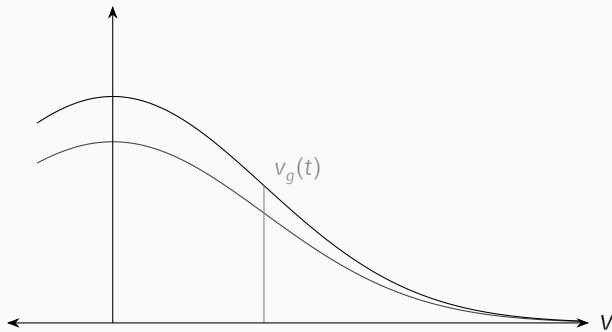
Let $A^p := \{g | \psi_g^p > 0\}$ denote the set of “active” genres with market price p .

Solving the Planner's Problem



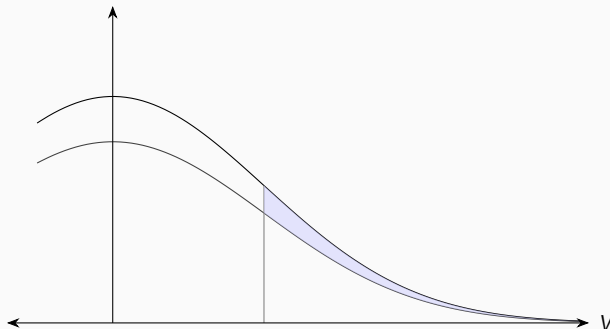
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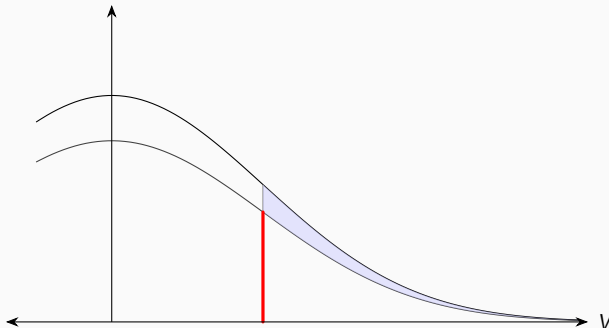
Solving the Planner's Problem



Consider the benefit from increasing v_g :

$$\int_{v_g(t)}^{\infty} \left(v \right) f(v) dv$$

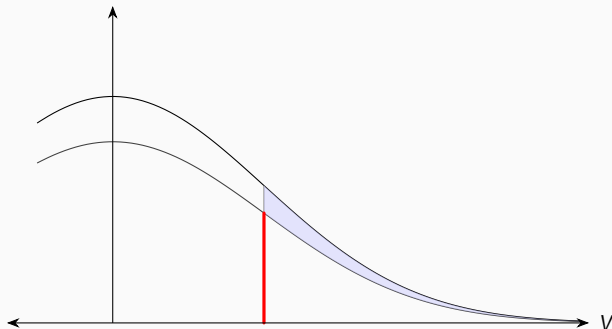
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Consider the benefit from increasing ψ_g :

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Solving the Planner's Problem



Consider the benefit from increasing ψ_g :

$$M(x) := \int_{\bar{F}^{-1}(x)}^{\infty} \left(v - \bar{F}^{-1}(x) \right) f(v) dv$$

Planner's Optimality

The planner hence sets

$$\int_0^\infty M\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right) dT_g(t)$$

equal across all active g .

Lemma

The socially optimal ψ^* is the unique allocation such that

$$\psi_g^* > 0 \Rightarrow g \in \operatorname{argmax}_{g \in G} \int_0^\infty M\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g^*}\right\}\right) dT_g(t)$$

Let $A^* := \{g | \psi_g^* > 0\}$. This is the set of “active” genres in the efficient allocation.

The Source of the Inefficiency

In the decentralized equilibrium, producers set

$$\int_0^{\infty} \Pi_p \left(\min \left\{ \bar{F}(p), \frac{t}{\psi_g} \right\} \right) dT_g(t)$$

equal. In the centralized (efficient) allocation, the planner sets equal

$$\int_0^{\infty} M \left(\min \left\{ \bar{F}(0), \frac{t}{\psi_g} \right\} \right) dT_g(t)$$

These functions are clearly not the same.

(Re)statement of Main Result

Theorem

Fix any $p \geq 0$.

- (i) If quality is realized ex-ante or pricing is competitive, there is no allocative inefficiency above p .

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- (ii) If quality is realized ex-post *and* pricing is fixed, and the platform's allocation is non-degenerate ($|A^*| > 1$), then there is generically allocative inefficiency: $\psi^P \neq \psi^*$.

Genericity is with respect to T , and is required because of edge-cases (e.g., if T_g is equal for all g).

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This allocative inefficiency is driven by the risk facing producers being different from the risk facing the platform.

Range: A measure of concentration

Recall that A^S is the set of active genres in market structure S : $A^S := \{g : \psi_g^S > 0\}$.

Definition

Given two market structures, S, S' , I say that S features *less range* than S' if $A^S \subseteq A^{S'}$.

Product range is a coarse measure of concentration: how many genres are produced in a market structure.

The nature of the failure: Product diversity

Lemma

Π_p is linear while M is strictly convex.

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Theorem

- (i) All competitive equilibria have less range than the efficient allocation: for all $p \geq 0$, $A^p \subseteq A^*$;

The nature of the failure: Product diversity

Lemma

Π_p is linear while M is strictly convex.

- The convexity of M pushes the planner to expand its range.

Theorem

- (i) All competitive equilibria have less range than the efficient allocation: for all $p \geq 0$, $A^p \subseteq A^*$;
- (ii) Range decreases in base price: A^p is shrinking in p .

Other Results

Proposition

Consider two genres g, g' with $T_{g'}$ a mean-preserving spread of T_g .

- (i) $\psi_g^p \geq \psi_{g'}^p$, for all p .
- (ii) There exists an $\bar{m}$ such that if the mass of producers is $m > \bar{m}$, $\psi_g^* \leq \psi_{g'}^*$.

Proposition

Fix a transformation $v \mapsto \tau(v)$ where τ is increasing with $\tau(0) = 0$. This induces distribution $\hat{F}(v) = F(\tau^{-1}(v))$. Assume that $\hat{F}$ has IHR.

- (i) If τ is concave, then $A^* \subseteq A_{\tau}^*$.
- (ii) If τ is convex, then $A^* \supseteq A_{\tau}^*$.
- (iii) $A^0 = A_{\tau}^0$.

- A monopolist producer subject to the fixed market price p will be more diverse than the decentralized equilibrium, and possibly more diverse than the efficient allocation.
- An integrated platform using genre-specific subscription pricing will be more diverse than optimal.
- A recommendation system can improve upon the decentralized equilibrium by committing to suppressing content from genres that are “too popular.”

I show how interacting frictions lead to inefficiencies in markets for cultural goods.

- In particular, I show that ex-ante quality uncertainty, together with pricing frictions, lead to an inefficient lack of diverse content.
- Either friction alone does not lead to inefficiency.
- In the paper, I explore how changing market structures interact with these frictions.

Competitive Pricing, Ex-ante Realization of Quality

- Firms observe their quality, choose their genre, observe competition, and then set prices.

Lemma

There exists a unique pricing equilibrium $p_g(v)$ such that $v - p_g(v)$ is increasing in v .

Proposition

The allocation of producers under competitive pricing and ex-ante realization of quality is efficient.

Competitive Pricing, Ex-post Realization of Quality

The timing is different here: producers choose their genre, observe their quality, and then set their price.

Lemma

For any genre g and mass of producers ψ_g , there exists a unique pricing equilibrium $p_g(v)$ such that $v - p_g(v)$ is increasing in v .

Proposition

The allocation of producers under competitive pricing and ex-post realization of quality is efficient.

Fixed Pricing, Ex-ante Realization of Quality

Consider a fixed price p . If producers know their quality they perfectly anticipate their demand from each genre.

- The highest quality producers will claim the highest demand.

Definition

- Let μ^p be the distribution of quality under the decentralized equilibrium.
- Let μ^* be the socially efficient distribution of producers when quality is realized ex-ante.

They are both unique for producers with positive demand ($v \geq p$).

Proposition

If quality is realized ex-ante, then there is no allocative inefficiency above the market price p : $\mu_g^p(B) = \mu_g^*(B)$ for all $g \in G$ and $B \subseteq (p, \infty)$.